Milestone 3 example:

	Number	Name	Bus 1	Bus 2
1	1	1	1.31 - j13.86	-1.31 + j13.88
2	2	2	-1.31 + j13.88	1.31 - j13.86

```
Ybus Matrix:
                    Bus 1              Bus 2
Bus 1  1.3081-13.8556j -1.3081+13.8756j
Bus 2 -1.3081+13.8756j  1.3081-13.8556j
```

Book example:

```
C:\Users\misie\PycharmProjects\ECE-1774-Project-2\.venv\Scripts\python.exe C:\Users\misie\Pycha
                 Bus 1             Bus 2             Bus 3              Bus 4             Bus 5
Bus 1  3.7290-49.7203j   0.0000+0.0000j   0.0000+0.0000j    0.0000+0.0000j -3.7290+49.7203j
Bus 2   0.0000+0.0000j  2.6783-28.4590j   0.0000+0.0000j   -0.8928+9.9197j -1.7855+19.8393j
Bus 3   0.0000+0.0000j   0.0000+0.0000j  7.4580-99.4406j  -7.4580+99.4406j   0.0000+0.0000j
Bus 4   0.0000+0.0000j  -0.8928+9.9197j -7.4580+99.4406j 11.9219-147.9589j -3.5711+39.6786j
Bus 5 -3.7290+49.7203j -1.7855+19.8393j   0.0000+0.0000j  -3.5711+39.6786j 9.0856-108.5782j
```

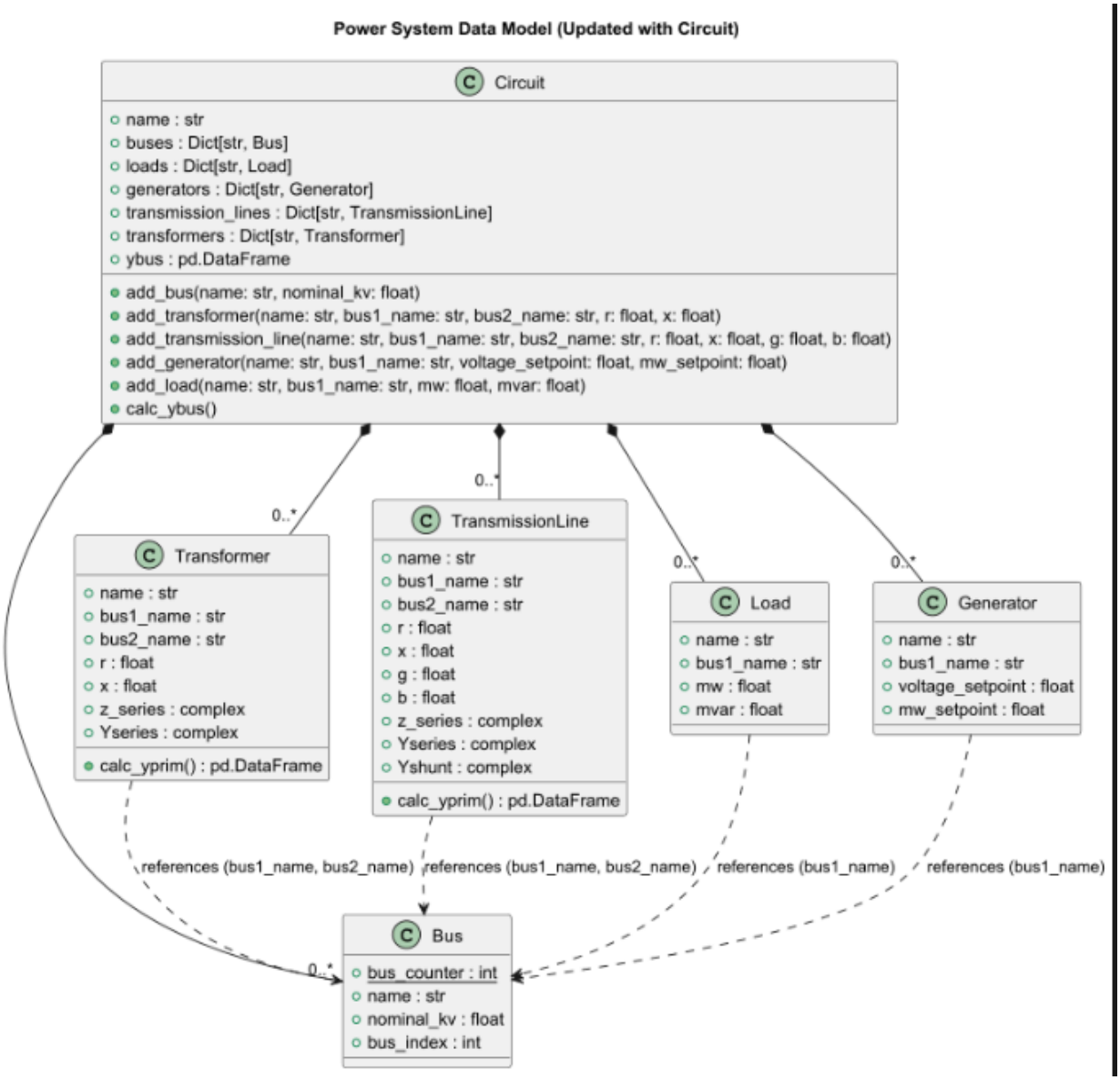
Y Bus (Bus Admittance Matrix) | Mismatches | Power Flow Jacobian

Records Geo Set Columns Options

Filter Advanced Bus Find... Remove

	Number	Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
1	1	One	3.73 - j49.72				-3.73 + j49.72
2	2	Two		2.68 - j28.46		-0.89 + j9.92	-1.79 + j19.84
3	3	Three			7.46 - j99.44	-7.46 + j99.44	
4	4	Four		-0.89 + j9.92	-7.46 + j99.44	11.92 - j147.96	-3.57 + j39.68
5	5	Five	-3.73 + j49.72	-1.79 + j19.84		-3.57 + j39.68	9.09 - j108.58

Search Search Now Options


Power System Data Model (Updated with Circuit)
Circuit
name : str
buses : Dict[str, Bus]
loads : Dict[str, Load]
generators : Dict[str, Generator]
transmission_lines : Dict[str, TransmissionLine]
transformers : Dict[str, Transformer]
ybus : pd.DataFrame
add_bus(name: str, nominal_kv: float)
add_transformer(name: str, bus1_name: str, bus2_name: str, r: float, x: float)
add_transmission_line(name: str, bus1_name: str, bus2_name: str, r: float, x: float, g: float, b: float)
add_generator(name: str, bus1_name: str, voltage_setpoint: float, mw_setpoint: float)
add_load(name: str, bus1_name: str, mw: float, mvar: float)
calc_ybus()
0..*
Transformer
name : str
bus1_name : str
bus2_name : str
r : float
x : float
z_series : complex
Yseries : complex
calc_yprim() : pd.DataFrame
TransmissionLine
name : str
bus1_name : str
bus2_name : str
r : float
x : float
g : float
b : float
z_series : complex
Yseries : complex
Yshunt : complex
calc_yprim() : pd.DataFrame
Load
name : str
bus1_name : str
mw : float
mvar : float
Generator
name : str
bus1_name : str
voltage_setpoint : float
mw_setpoint : float
references (bus1_name, bus2_name)
references (bus1_name, bus2_name)
references (bus1_name)
references (bus1_name)
Bus
bus_counter : int
name : str
nominal_kv : float
bus_index : int